

JAVA PROGRAMMING

MODULE 1

1. List any four features of Java.
2. What is byte code in Java?
3. Define JVM and mention its purpose.
4. Differentiate between JDK and JRE.
5. List the different types of primitive data types in Java.
6. What is type casting in Java?
7. Define a variable in Java. Give an example.
8. List the different types of operators available in Java.
9. What are control statements in Java? Name their types.
10. Define looping statements in Java and list the different types of loops.
11. Design a Java program to demonstrate the use of different data types, variables, and type casting. Explain how type conversion takes place in your program.
12. Develop a Java program using control statements to check whether a given number is positive, negative, or zero and display an appropriate message.
13. Create a Java program to generate the multiplication table of a given number using looping statements.
14. Design a Java program to store and process student marks using an array. The program should calculate the total and average marks.
15. Construct a Java program to perform string operations such as finding the length, reversing a string, and comparing two strings.
16. Develop a Java class named Student with suitable data members and methods to store and display student details. Create objects to access the class members.
17. Design a Java program to demonstrate the use of access specifiers (public, private, and protected) in a class.
18. Create a Java program to demonstrate the use of the this keyword for distinguishing instance variables and method parameters.
19. Develop a Java program using a static variable and static method to maintain common information shared by multiple objects.
20. Design a Java program to demonstrate the concept of nested or inner classes and explain how objects of these classes are created.

21. A student wants to develop a Java program that calculates the average marks of five subjects. Write a Java program using appropriate data types, variables, operators, and type conversion to display the average mark.
22. A company needs a simple employee management program. Create a Java class Employee with suitable data members and methods. Demonstrate the creation of objects and access the members using appropriate access specifiers.
23. Develop a Java program to check whether a given number is prime or not using suitable control statements and looping statements.
24. A college wants to store the marks of students in an array and find the highest mark. Write a Java program using arrays and looping statements to perform this task.
25. Write a Java program to demonstrate the use of the this keyword by creating a class Student with instance variables and constructor. Use this keyword to differentiate between local and instance variables.
26. A banking application requires storing account details. Create a Java class BankAccount using a static variable to maintain the bank name and instance variables for account details. Create objects and display the information.
27. Develop a Java program to count the number of vowels and consonants in a given string. Use appropriate String methods and looping statements.
28. Write a Java program to demonstrate type casting by converting a floating-point value into an integer and displaying the result. Explain the effect of type conversion on the data.
29. A software developer wants to create a utility class inside another class for handling calculations. Write a Java program demonstrating the use of a nested class or inner class.
30. A programmer develops a Java application and wants to understand how the source code is executed. Explain the role of JDK, JRE, JVM, and Byte Code by describing the steps involved in executing a simple Java program.

31. Define method overloading in Java.
32. What is a constructor? Mention any one characteristic of constructors in Java.
33. Define method overriding in Java.
34. What is inheritance? Name any two types of inheritance supported in Java.
35. What is Dynamic Method Dispatch in Java?
36. Define an abstract class in Java.
37. What is an interface in Java?
38. Write the syntax for defining an interface in Java.
39. What is a package in Java? How do you declare a package?
40. What are exceptions in Java? List any two types of exceptions.
41. What is the purpose of the import statement in Java?
42. Define a sub-package in Java.
43. What is the use of the try-catch block in Java?
44. Differentiate between throw and throws keywords in Java.
45. What is the purpose of the finally block in Java?
46. Design a Java program to create a class Student with multiple constructors to initialize student details using different sets of parameters. Demonstrate constructor overloading.
47. Develop a Java program to create a superclass Employee and a subclass Manager. Implement method overriding to display different details of employees.
48. Create a Java program to demonstrate dynamic method dispatch by using a superclass reference to call overridden methods of different subclasses.
49. Design a Java application using an abstract class Shape with abstract methods to calculate area. Create subclasses such as Circle and Rectangle to implement the methods.
50. Develop a Java program to define an interface Payment with required methods and implement it using different classes such as CreditCard and UPI.
51. Create a Java package named studentinfo containing a class to store student details. Write another Java program to import and use this package.
52. Design a Java program using sub-packages to organize classes related to college.student and college.staff. Access the classes from the main program.
53. Develop a Java program to handle arithmetic exceptions using try-catch blocks. Create suitable exception handling statements to prevent program termination.
54. Create a Java program to validate user input and use the throw statement to generate a user-defined exception when invalid data is entered.

55. Design a Java program that demonstrates the use of throws and finally statements while handling file or arithmetic exceptions.
56. Develop a Java program to demonstrate method overloading by creating different versions of a method to calculate the area of a circle, rectangle, and triangle. Explain how the compiler identifies the correct method.
57. Write a Java program to create a class Student using constructors. Implement default and parameterized constructors to initialize student details and display the information.
58. Create a Java program to demonstrate inheritance and method overriding. Define a parent class Vehicle and child classes Car and Bike that override a display() method.
59. Develop a Java program to demonstrate Dynamic Method Dispatch using a superclass reference to call overridden methods of different subclass objects.
60. Design a Java program using an abstract class Shape with an abstract method area(). Implement the method in Circle and Rectangle classes to calculate their areas.
61. Create a Java interface named Payment with a method pay(). Implement this interface in classes CreditCard and UPI to demonstrate multiple implementations.
62. Develop a Java program to create a user-defined package containing a class Calculator. Import the package into another program and perform arithmetic operations.
63. Write a Java program to demonstrate the use of sub-packages. Create a package structure and access a class from a sub-package using import statements.
64. Develop a Java program to handle arithmetic exceptions. Use try-catch, throw, throws, and finally blocks to handle division by zero and display appropriate messages.
65. Create a Java application for a bank account system that uses inheritance, method overriding, interfaces, and exception handling to perform deposit and withdrawal operations safely.